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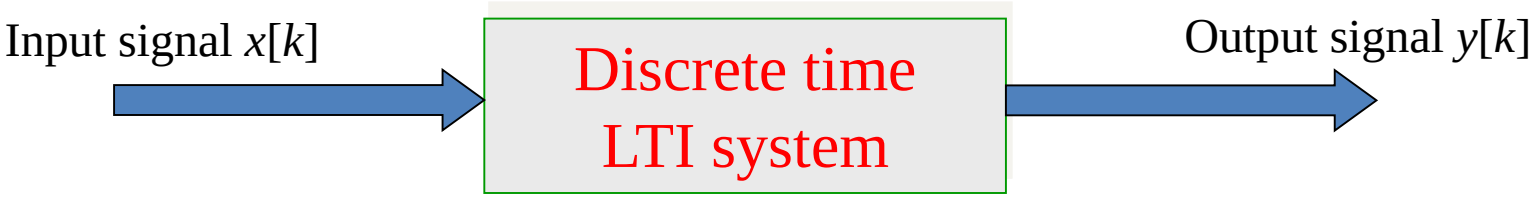
Time domain analysis of discrete LTI system responses



- Iterative method
- Method based on solving the difference equation of the linear constant coefficient
- Methods based on zero input responses versus zero state responses

1. Iterative method

[Example] The known difference equation is $y[k]-0.5y[k-1]=u[k]$, $y[-1]=1$. Find the output signal.

Solution: Write the difference equation as $y[k]=u[k]+0.5y[k-1]$

Assume the initial state $\Rightarrow$ $y[0]=u[0]+0.5y[-1]=1+0.5\times 1=1.5$

And so on $y[1]=u[1]+0.5y[0]=1+0.5\times 1.5=1.75$

$$y[2]=u[2]+0.5y[1]=1+0.5\times 1.75=1.875$$

$$\vdots$$

n initial states are known { $y[-1], y[-2], y[-2],\cdots, y[-n]$ } and inputs, the outputs of the system iterated out by the differential equations, are called iterative methods.

Pros: Simple and straightforward, suitable for computer computing; Cons: It is difficult to get a closed form of solution

2. Method based on solving the difference equation of linear constant coefficients

Describes a discrete LTI system using a linear constant coefficient difference equation

$$\sum_{i=0}^{n} a_i y[k-i] = \sum_{j=0}^{m} b_j x[k-j]$$

The full solution of the difference equation consists of a homogeneous solution of $y_h[k]$ and a special solution of $y_p[k]$.

$$y[k] = y_h[k] + y_p[k]$$

✓ The form of the homogeneous solution $y_h[k]$ is determined by the corresponding feature root of the difference equation

✓ The form of the special solution $y_p[k]$ is determined by the form of the excitation signal on the right side of the equation

The full solution of the differential equation is the output response of the system.

2. Method based on solving the difference equation of linear constant coefficients

[Example] The difference equation that describes a discrete LTI system is known

$$y[k]-5y[k-1]+6y[k-2]=x[k],\quad k\geq 0$$

Initial conditions $y[0]=0$, $y[1]=-1$, input $x[k]=4^k u[k]$, find full solution $y[k]$.

Solution: (1) Homogeneous solution of the difference equation $y_h[k]$

$y[k]-5y[k-1]+6y[k-2]=0$ forms of homogeneous solutions

The characteristic equation is $r^2-5r+6=0$

The feature root is $r_1=2, r_2=3$

Homogeneous solution $y_h[k]$ $\quad y_h[k]=C_1 2^k+C_2 3^k,\quad k\geq 0$

2. Method based on solving the difference equation of linear constant coefficients

[Example] The difference equation that describes a discrete LTI system is known

$$y[k]-5y[k-1]+6y[k-2]=x[k],\quad k\geq 0$$

Initial conditions $y[0]=0$, $y[1]=-1$, input $x[k]=4^k u[k]$, find full solution $y[k]$.

solution: (2) Solve the difference $y[k]-5y[k-1]+6y[k-2]=x[k]$ of the special solution $y_p[k]$

By entering $x[k]$, let the special solution of the equation be

$$y_p[k]=A\cdot 4^k,\quad k\geq 0$$

Bring the special solution into the original difference equation to find the coefficients to be determined $A=8$。

2. Method based on solving the difference equation of linear constant coefficients

[Example] The difference equation that describes a discrete LTI system is known

$$y[k]-5y[k-1]+6y[k-2]=x[k],\quad k\geq 0$$

Initial conditions $y[0]=0$, $y[1]=-1$, input $x[k]=4^k u[k]$, find full solution $y[k]$.

solution: (3) Find the full solution of the difference equation

$$y[k]=y_{\mathrm{h}}[k]+y_{\mathrm{p}}[k]=C_1 2^k+C_2 3^k+8\cdot 4^k,\quad k\geq 0$$

$$y[0]=C_1+C_2+8=0$$

$$y[1]=2C_1+3C_2+32=-1 \quad \Longrightarrow \quad C_1=9,\ C_2=-17$$

$$y[k]=9\cdot 2^k-17\cdot 3^k+8\cdot 4^k,\quad k\geq 0$$

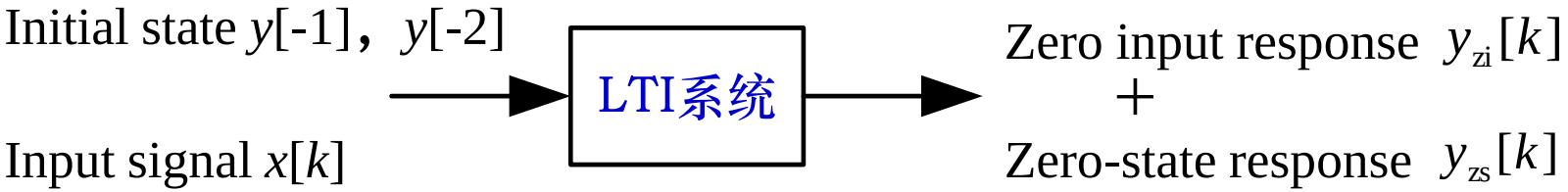
2. Method based on solving the difference equation of linear constant coefficients

The inadequacies of the method based on solving the difference equation of the linear constant coefficients

- If the input signal changes, it must be fully resolved.
- If the initial conditions change, they must all be resolved.
- If the stimulus term of the difference equation is more complex, it is difficult to deal with.
- The physical concept of the response of a discrete LTI system cannot be represented.

3. Methods based on zero input responses and zero-state responses

According to the linear characteristics of the discrete LTI system, the system response is regarded as a superposition of the response produced by the initial state and the input signal acting separately on the system.



Fully responsive $y[k]$ **= Zero input response** $y_{zi}[k]$ **+ Zero status response** $y_{zs}[k]$

3. Methods based on zero input responses and zero-state responses

Zero input response

The input signal is zero, and the response produced by acting solely on the initial state of the system is called the zero input response, which is denoted $y_{zi}[k]$

Describes the LTI system using a constant-coefficient linear difference equation

$$\sum_{i=0}^{n} a_i y[k-i] = \sum_{j=0}^{m} b_j x[k-j]$$

When entering $x[k]=0$

$$\sum_{i=0}^{n} a_i y[k-i] = 0$$

The above equation is homogeneous, so the zero input response has the form of a homogeneous solution

3. Methods based on zero input responses and zero-state responses

Zero-state response

The initial state is zero, and the response is generated only by the external signal x[k] of the system. Called the zero-state response of the system, it is denoted by $y_{zs}[k]$.

Zero-state response analysis method

(1) Represents the input signal as a linear combination of unit pulse train

(2) Find the zero-state response of the unit pulse train acting on the system—the unit impulse response $h[k]$

(3) Using the LTI system feature, the zero-state response of the system under input x[k] excitation can be obtained $y_{zs}[k]$。

3. Methods based on zero input responses and zero-state responses

Zero-state response

$\delta[k]$ → 离散LTI系统 → $h[k]$ Unit impulse response

Non-time-varying characteristics: $\delta[k-n]$ → 离散LTI系统 → $h[k-n]$

Uniformity: $x[n]\delta[k-n]$ → 离散LTI系统 → $x[n]h[k-n]$

Overlay: $\sum_{n=-\infty}^{\infty} x[n]\delta[k-n]$ → 离散LTI系统 → $\sum_{n=-\infty}^{\infty} x[n]h[k-n]$

$x[k]$

$y_{zs}[k]=x[k]*h[k]$

3. Methods based on zero input responses and zero-state responses

System Response $y[k]$ = Zero Input Response $y_{zi}[k]$ + Zero Status Response $y_{zs}[k]$

Solve the homogeneous difference equation

$$y_{zs}[k] = x[k] * h[k]$$

※ Solution of zero input response

※ Unit impulse response solution

※ Solve for zero-state response

Thanks

Some of the materials cited in this course are the teaching accumulation of the main lecturer for many years, which come from a variety of media and the exchanges of colleagues, peers and friends, and it is difficult to indicate the source one by one, so I hereby explain and thank you!